Real-World Problems and Significant Digits

Teacher's Edition

The Period

MA.8.NSO.1.6 Solve real-world problems involving operations with numbers expressed in scientific notation.

Must Solve real-world problems with numbers in scientific notation, and recognise the importance of significant digits when physical measurements are involved.

Must not The within-2 exponent limit still applies to addition and subtraction here. Significant digits limit a reported answer; they never change the arithmetic, and a count is never a measurement.

Target I can solve a real-world problem with numbers in scientific notation and report the answer with the right number of significant digits.

MTR **MTR.6.1** — Boards 2, 6 and 7 · **MTR.7.1** — Every problem from Notes II on · **MTR.4.1** — Board 6

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	12 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. A measurement's last digit is uncertain. That is what the whole rule is protecting.
2. Products and quotients count digits. Sums and differences count places. Two rules, and it matters which one you reach for.
3. Do all the arithmetic first and round once, at the end. Rounding in the middle loses digits the arithmetic still needed.

Slide by Slide

1. **Real-World Problems and Significant Digits.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q3 and Q4 are today's two tools. *1 retrieves 4.02–03 — multiply the coefficients, add the exponents, then finish the answer (yesterday). 2 retrieves 4.01 — rewrite one number over the other's power, then subtract the coefficients (last week). 3 retrieves 3.09 — small numbers; note that the zero between the 9 and the 2 stays (last unit). 4 retrieves MA.5.NSO.1.5 — rounding to a named place; every answer today ends with one of these (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 1.2×10^{4} 2. 5.89×10^{7} 3. 9.02×10^{-3} 4. 4.74
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the table.** Scientific notation makes this easy. In 5.2 times ten to the fourth, every digit you can see is significant — that is one of the reasons the notation exists.
6. **Notes.** Two significant digits in, two significant digits out. The answer is never better than the measurement it came from.
7. **Notes — Copy all three lines.** Products count digits. Sums count places. They are two different rules and it matters which one you reach for.
8. **Example 1 — A product of two measurements.** SPOKEN ONLY.
9. **Example 1 — Worked.** Multiply in full — 10,750 — and round once.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same shape, the counts the other way round.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A difference of two measurements.** SPOKEN ONLY.
13. **Example 2 — Worked.** Subtract in full, then stop at the leftmost last-digit place.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 3,850 stops at the tens place and 274 stops at the ones, so the answer stops at the tens.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **Board 1.** How many significant digits are in 0.00470 ? → ***3*** *2 — the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]; 5 — the leading zeros were counted [B1G-M 8.NSO.1.6: leading zeros are never significant]*
17. **Board 1.**
18. **Board 2.** A pond's surface measures 1.4×10^{2} m by 8.5×10^{1} m. The pond is 3 meters deep. Find the surface area. Round to the correct number of significant digits. → ***1.2×10^{4} square meters*** *The depth of 3 meters is not needed. anything built from the depth — the unneeded figure was used [an area needs two dimensions]; 1.19×10^{4} — the arithmetic without the rounding [8.NSO.1.6 clarification]*
19. **Board 2.**
20. **Board 3.** Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^{5}. How should the answer be written? → ***7.41×10^{5}*** *7.4×10^{5} — two digits, the wrong one of the two counts [8.NSO.1.6: the answer takes the smaller count, which is 3]; 7.4126×10^{5} — no rounding at all*
21. **Board 3.**

22–23. Board 4. $(2.5 \times 10^3) + (4.71 \times 10^2)$ meters, both measured. Write the sum with the correct precision. → ***3.0 × 10³ m*** *2.971 × 10³ — the arithmetic without the precision rule [8.NSO.1.6 clarification]; 2.9 × 10³ — 2,971 rounded down instead of up [rounding to the hundreds place]*

24–25. Board 5. How many significant digits are in 0.03080? → ***A — 4*** *(B) only the nonzero digits were counted [B1G-M 8.NSO.1.6: a zero between nonzero digits is always significant]; (C) one of the leading zeros was counted [B1G-M 8.NSO.1.6: leading zeros are never significant]; (D) the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm. Which is the area, to the correct significant digits? → ***B — 2.6 × 10⁵ cm²*** *(A) the exact product, not reduced, and 26.1 is not less than 10 — two mistakes [B1G-M 8.NSO.1.5 misconception: leaving 12×10^9 instead of 1.2×10^{10}]; (C) the exponent was not raised when the coefficient was repaired [8.NSO.1.5: point one place left, exponent up one]; (D) the exact product, not reduced to the two significant digits the measurements carry [8.NSO.1.6 clarification] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A car travels 4.20×10^2 km in 3.0 hours. What is its average speed, to the right number of significant digits? → ***1.4 × 10² km/h*** *140.0 — four digits claimed from a two-digit measurement [8.NSO.1.6 clarification]; 1.4×10^1 — the exponent is one too low*

30–31. Board 8. A measured length times a measured width gives an area of 3.6×10^3 m². The length measured 9.0×10^1 m. What did the width measure, to two significant digits? → ***4.0 × 10¹ m*** *4×10^1 — one significant digit where two were asked for [8.NSO.1.6: a trailing zero after a decimal point carries the second digit]; 4.0×10^2 — the exponent is one too high*

32–33. Board 9. Ellis says that because 9,145 has 4 significant digits and 4.8 has 2, the sum 9,145 + 4.8 must be written with 2 significant digits, as 9.1×10^3. Explain why he is wrong, and give the correct sum. → ***9,150 = 9.150 × 10³*** *'he used the wrong rule' — the right rule is not stated [the item asks for an explanation]; 9,149.8 — no rounding at all, which claims a tenth that 9,145 never measured*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Questions 1–4 are the four counting rules, one each: a middle zero, leading zeros, trailing zeros with a decimal point, trailing zeros without one. 5–8 are the four operations in order — a product, a quotient, a sum, a difference — so the two rules alternate; 7 is the one that shows a sum keeping more significant digits than either measurement — three, from two two-digit measurements. 9 and 10 are real quantities, and 10 is built on the count-versus-measurement distinction. The pattern breaks at 11, where the arithmetic is right and the answer is wrong. 12 is the diagnostic multiple choice, stacking the coefficient repair on the rounding. The Additional sheet mirrors each position, and its question 11 is the case where the two rules agree — which is why the wrong one survives.

Misconceptions to Watch

Board 1. 2 — the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]

Board 2. anything built from the depth — the unneeded figure was used [an area needs two dimensions]

Board 3. 7.4×10^5 — two digits, the wrong one of the two counts [8.NSO.1.6: the answer takes the smaller count, which is 3]

Board 4. 2.971×10^3 — the arithmetic without the precision rule [8.NSO.1.6 clarification]

Board 5. only the nonzero digits were counted [B1G-M 8.NSO.1.6: a zero between nonzero digits is always significant]

Board 6. the exact product, not reduced, and 26.1 is not less than 10 — two mistakes [B1G-M 8.NSO.1.5 misconception: leaving 12×10^9 instead of 1.2×10^{10}]

Board 7. 140.0 — four digits claimed from a two-digit measurement [8.NSO.1.6 clarification]

Board 8. 4×10^1 — one significant digit where two were asked for [8.NSO.1.6: a trailing zero after a decimal point carries the second digit]

Board 9. 'he used the wrong rule' — the right rule is not stated [the item asks for an explanation]

Materials Whiteboards and markers. Calculators are allowed on everything and are genuinely useful today, because the point of the lesson is what to do with a long display rather than how to produce one.